Practical Digital System Design　　Workstation Usage Process Worksheet

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Name : 尋以諾

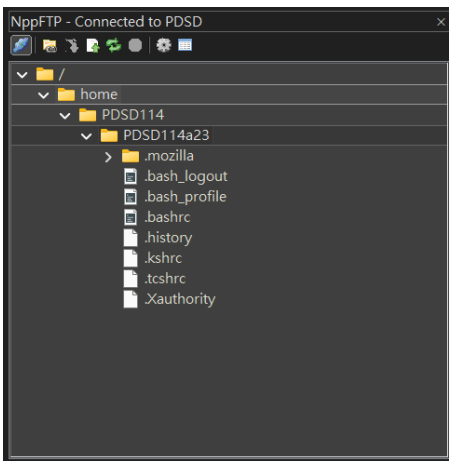
Based on the workstation usage process introduction in handout. Please answer the following questions.

【Phase 1】: Setup and connection to the workstation

1. According to the VPN usage instructions from the Office of Library and Information Services, after completing the setup of Invanti Secure Access Client, can the program icon be seen in the taskbar on your local machine, and does it show as connected? __Y__



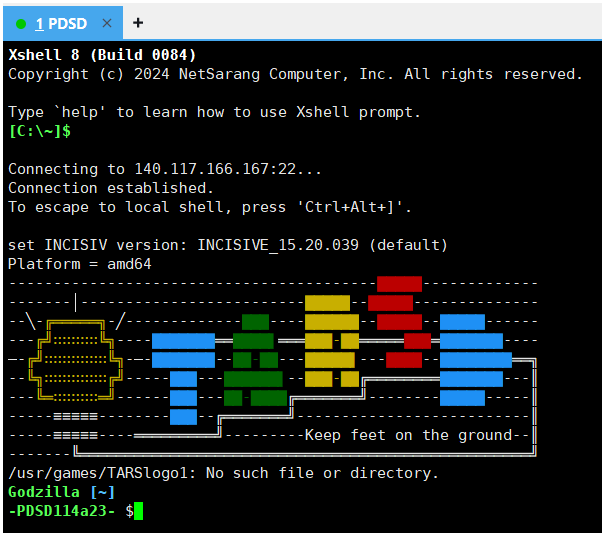
2. According to pages 5 to 9 of the “03-I SOC Server Overview” handout, after completing the Notepad++ setup, can you see the folders and files of your workstation account in the NppFTP Window? __Y__



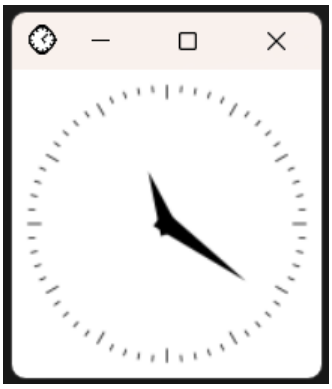
3. 3. According to pages 10 to 14 of the “03-I SOC Server Overview” handout, after completing the Xming setup, can the program icon be seen in the taskbar on your local machine? __Y__



4. According to pages 15 to 22 of the SOC Overview handout, after completing the Xshell setup, can you see the TARS logo and your workstation account in the terminal? __Y__



5. According to page 23 of the “03-I SOC Server Overview” handout, after entering the “xclock &” command in the terminal, can you see a window displaying the clock icon? __Y__



6. According to pages 24 to 26 of the “03-I SOC Server Overview” handout, after completing the FileZilla setup, can you see the folders and files of your workstation account on the remote site? __Y__



【Phase 2】: Simulation and Verification

1. Download the EX1.zip file from the cyber university, extract it, and upload it to your workstation account using FileZilla. Can you see the EX1 folder on the remote site? __Y__



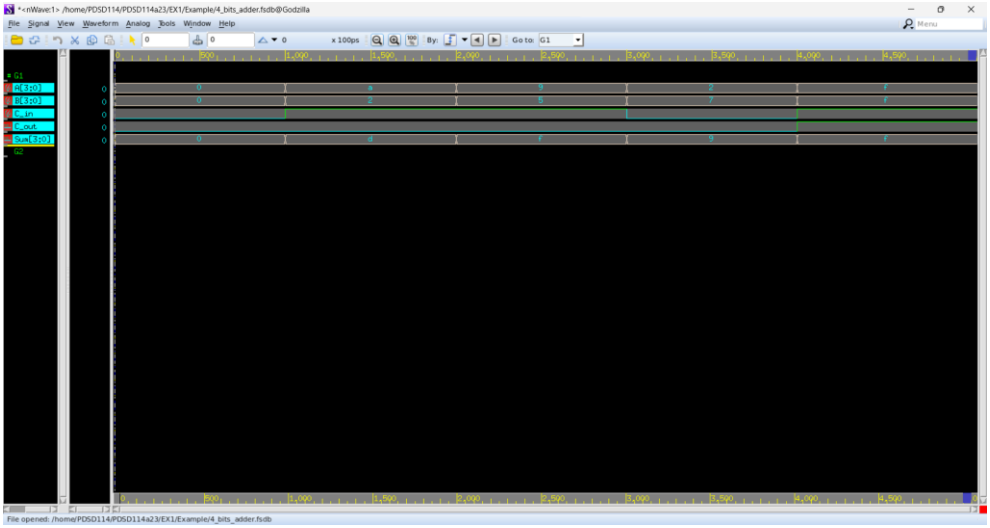
2. According to page 9 of the “03-II_Operation in Workstation” handout, after entering the command ncverilog four_bits_RCA.v -c in the terminal, does it check if the syntax is correct? __Y__

```
-PDSD114a23- $cd /home/PDSD114/PDSD114a23/EX1/Example/
Godzilla [~/EX1/Example]
-PDSD114a23- $ll
total 16
drwxr-xr-x. 2 PDSD114a23 PDSD114 4096 Sep 17 11:25 dat/
-rw-r--r--. 1 PDSD114a23 PDSD114  244 Sep 17 11:30 four_bits_RCA.v
-rw-r--r--. 1 PDSD114a23 PDSD114 4078 Sep 17 11:25 testfixture_auto.v
-rw-r--r--. 1 PDSD114a23 PDSD114  947 Sep 17 11:25 testfixture.v
Godzilla [~/EX1/Example]
-PDSD114a23- $ncverilog four_bits_RCA.v -c
ncverilog(64): 15.20-s039: (c) Copyright 1995-2017 Cadence Design Systems, Inc.
file: four_bits_RCA.v
        module worklib.four_bits_RCA:v
                errors: 0, warnings: 0
                Caching library 'worklib' ....... Done
        Elaborating the design hierarchy:
        Building instance overlay tables: ..................... Done
        Generating native compiled code:
                worklib.four_bits_RCA:v <0x389d7681>
                        streams:   0, words:     0
        Building instance specific data structures.
        Loading native compiled code:     ..................... Done
        Design hierarchy summary:
                                Instances  Unique
                Modules:            1       1
                Scalar wires:       1       -
                Vectored wires:     2       -
                Cont. assignments:  0       2
        Writing initial simulation snapshot: worklib.four_bits_RCA:v
Godzilla [~/EX1/Example]
-PDSD114a23- $
```

3. According to page 11 of the “03-II_Operation in Workstation” handout, after entering the command ncverilog testfixture.v +access+r in the terminal, can you see the simulation results? __Y__

```
-PDSD114a23- $ncverilog testfixture.v +access+r
ncverilog(64): 15.20-s039: (c) Copyright 1995-2017 Cadence Design Systems, Inc.
Loading snapshot worklib.textfixture:v ..................... Done
*Verdi* Loading libsscore_ius152.so
ncsim> source /usr/cad/cadence/INCISIV/INCISIVE_15.20.039/tools/inca/files/ncsimrc
ncsim> run
FSDB Dumper for IUS, Release Verdi_Q-2020.03, Linux x86_64/64bit, 02/09/2020
(C) 1996 - 2020 by Synopsys, Inc.
*Verdi* : Create FSDB file '4_bits_adder.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : End of traversing.
                   0 Sum= 0 A= 0 B= 0 C_in=0 C_out=0
                  10 Sum=13 A=10 B= 2 C_in=1 C_out=0
                  20 Sum=15 A= 9 B= 5 C_in=1 C_out=0
                  30 Sum= 9 A= 2 B= 7 C_in=0 C_out=0
                  40 Sum=15 A=15 B=15 C_in=1 C_out=1
Simulation complete via $finish(1) at time 500 NS + 0
./testfixture.v:28 #10 $finish;
ncsim> exit
Godzilla [~/EX1/Example]
-PDSD114a23- $
```

4. According to pages 14 to 18 of the “03-II_Operation in Workstation” handout, after completing the relevant operations, can you see the waveform simulation results? __Y__



5. Modify Pattern.dat and Expected.dat based on the input values to be tested (C_in, A, B) and the expected output results (C_out, Sum). After entering the command ncverilog testfixture_auto.v +access+r in the terminal, can you see the Totoro icon? __Y__

```
============================ Simulation START !! ============================

                    Pattern01:  ** Correct!! **
                    Pattern02:  ** Correct!! **
                    Pattern03:  ** Correct!! **
                    Pattern04:  ** Correct!! **
                    Pattern05:  ** Correct!! **

============================ Simulation Finish ! ============================

                ****************************
                **                        **       /|__/|
                **    Congratulations !!  **      / 0,0  |
                **                        **    /_____   |
                **  Simulation Complete!! **   /^ ^ ^ \  |
                **                        **  |^ ^ ^ ^ |w|
                ****************************   \m___m__|_|

Simulation complete via $finish(1) at time 550 NS + 0
./testfixture_auto.v:131        #5 $finish;
ncsim> exit
Godzilla [~/EX1/Example]
-PDSD114a23- $
```

【File Description】: EX1.zip

Folder name	File name		說明
Example	dat	Expected.dat	Output comparison data, which will be used in the automated Testbench
		Pattern.dat	Input test data, which will be used in the automated Testbench
	four_bits_RCA.v		4-bit Adder Circuit

	testfixture.v	Manual Testbench
	testfixture_auto.v	Automated Testbench

【Notes】：

Each person must submit a PDF file of this learning task to the EX1 folder on the FTP, with the submission deadline being before 9:00 on September 24, 2025 (Wednesday).